

The Endpoint Diameter Covering Number in Cwikel–Levy’s Example is Infinite

Packet for arXiv:0810.4240

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Source Question

The source is Michael Cwikel and Eliahu Levy, *Estimates for covering numbers in Schauder’s theorem about adjoints of compact operators*, arXiv:0810.4240. In Section 3.2 they construct an example for which $N_A(1/2) < \infty$ and $N_B^\Delta(\rho) = \infty$ for every $0 < \rho < 1/2$. They explicitly leave open the endpoint:

in this case equality holds in (8).

3.2. The statement (ii) in Theorem 4 is very close to best possible.

Our example in this subsection addresses the statement (ii) in Theorem 4. It is natural to ask whether (ii) might remain true if we replace ϵ in the requirement $\rho > \epsilon$ by some smaller number. In order to show that (ii) cannot be sharpened in such a way, we will give an example of sets A and B and a function $h : A \times B \rightarrow \mathbb{R}$ satisfying the hypotheses of Theorem 4, such that, when $\epsilon = 1/2$, we have $N_A(\epsilon) < \infty$, but $N_B^\Delta(\rho) = \infty$ for all $\rho \in (0, \epsilon)$. (Since we have not yet determined whether or not $N_B^\Delta(1/2)$ is finite, this does not yet settle the question of whether or not $N_A(\epsilon) < \infty$ always implies that $N_B^\Delta(\epsilon) < \infty$.)

In view of (6), statement (ii) of Theorem 4 implies the following similar statement, expressed solely in terms of intrinsic covering numbers:

$$(31) \quad \text{If } N_A(\epsilon) < \infty \text{ then } N_B(\rho) < \infty \text{ for all } \rho > 2\epsilon .$$

$$(32) \quad \min \{d_B(b, b_j) : 1 \leq j \leq m\} \leq \rho \text{ for all } b \in B.$$

Since each of the sequences b_j is finitely supported, there exists an integer q such that $\beta_{j,q} = 0$ for $j = 1, 2, \dots, m$. Now let $b = \{\beta_n\}_{n \in \mathbb{N}}$ be the sequence for which $\beta_q = 1$ and $\beta_n = 0$ for all $n \neq q$. Then $d_B(b, b_j) \geq 1$ for all j , which contradicts (32). Thus indeed, for each $\rho \in (0, 1)$, we have $N_B(\rho) = \infty$, and consequently (cf. (6)) also $N_B^\Delta(\rho/2) = \infty$. In other words we also have $N_B^\Delta(\rho) = \infty$ for each $\rho \in (0, 1/2)$. This shows that the covering numbers N_A , N_B^Δ and N_B have all the properties which were promised at the beginning of this subsection.

As indicated above, we would like to know whether or not $N_B^\Delta(1/2) < \infty$.

4. APPENDIX: FURTHER DETAILS ABOUT THE CASE WHERE A IS ABSOLUTELY

Answer

For the same example, $N_B^\Delta(1/2) = \infty$. Hence the implication

$$N_A(\epsilon) < \infty \implies N_B^\Delta(\epsilon) < \infty$$

fails in general. The strict inequality $\rho > \epsilon$ in Theorem 4(ii) of the source cannot be replaced by $\rho \geq \epsilon$.

The Source Example

The source takes $A = [0, 1]^\mathbb{N}$ and lets B be the real finitely supported ℓ_1 -unit ball:

$$B = \left\{ b = (\beta_n)_{n \in \mathbb{N}} : b \text{ is finitely supported and } \sum_{n=1}^{\infty} |\beta_n| \leq 1 \right\}.$$

For $a = (\alpha_n) \in A$ and $b = (\beta_n) \in B$, it sets

$$h(a, b) = \sum_{n=1}^{\infty} \alpha_n \beta_n.$$

The induced semimetric on B is

$$d_B(b, b') = \max \left\{ \sum_{n=1}^{\infty} (\beta_n - \beta'_n)_+, \sum_{n=1}^{\infty} (\beta'_n - \beta_n)_+ \right\}.$$

The diameter covering number $N_B^\Delta(\rho)$ is the least number of subsets needed to cover B , each subset having d_B -diameter at most 2ρ .

Proof Intuition

The source already proves that no finite intrinsic cover works below radius 1. At the diameter end-point $\rho = 1/2$, sets in a diameter cover may have diameter 1, so the old one-coordinate obstruction no longer separates points: the unit vectors e_i are exactly distance 1 apart.

The small extra trick is to use signed two-point vectors with different positive masses. If two such vectors have disjoint supports, the positive variation in one direction and the negative variation in the other direction add up to $1 + |p - q|$, which is strictly larger than 1 whenever the masses are distinct. Thus one obtains infinitely many points no two of which can lie in a single diameter-1 set.

Theorem 1. *In the example of Section 3.2 of Cwikel–Levy, $N_B^\Delta(1/2) = \infty$.*

Proof. Choose a sequence of pairwise distinct numbers $p_k \in (0, 1)$, for instance $p_k = 1/(k+2)$. Let e_j denote the standard unit sequence and define

$$b^{(k)} := p_k e_{2k} - (1 - p_k) e_{2k+1} \quad (k \geq 1).$$

Each $b^{(k)}$ is finitely supported and

$$\|b^{(k)}\|_1 = p_k + (1 - p_k) = 1,$$

so $b^{(k)} \in B$.

Fix $j \neq k$. Since the supports of $b^{(j)}$ and $b^{(k)}$ are disjoint, the positive part of $b^{(j)} - b^{(k)}$ has total mass

$$p_j + (1 - p_k) = 1 + p_j - p_k,$$

while the negative part has total mass

$$(1 - p_j) + p_k = 1 - p_j + p_k.$$

Therefore

$$d_B(b^{(j)}, b^{(k)}) = \max\{1 + p_j - p_k, 1 - p_j + p_k\} = 1 + |p_j - p_k| > 1.$$

A set of d_B -diameter at most 1 can contain at most one element of the infinite family $\{b^{(k)} : k \geq 1\}$. Hence B cannot be covered by finitely many subsets of d_B -diameter at most 1. By the definition of diameter covering number, this says exactly that

$$N_B^\Delta(1/2) = \infty.$$

□

Verification and Scope

This proof answers only the endpoint diameter-covering question left in Section 3.2 of the source paper. It does not change the source theorem; rather, it confirms that the source example also rules out replacing $\rho > \epsilon$ by $\rho \geq \epsilon$ in Theorem 4(ii).

Bounded novelty check: local run indexes were searched for arXiv:0810.4240 and the main endpoint terminology. Web searches on 2026-06-20 for exact variants of the endpoint question found no later explicit answer beyond the source arXiv record.

References

- [1] M. Cwikel and E. Levy, *Estimates for covering numbers in Schauder's theorem about adjoints of compact operators*, arXiv:0810.4240.